



Estd:1946

THE NATIONAL INSTITUTE OF ENGINEERING

(An Autonomous Institution under VTU, Belagavi)

Manandavadi Road, Mysuru – 570 008.

Department of Computer Science & Engineering (AIML)

Course Plan for the term ODD 2025-26

Course Instructor: Mr. Mohammed Adnan

Course Title: Collaborative Development and DevOps

LTPS: 42:0:28

Sec: CSE “B”

Course Code: 1BCS303

Semester: III

LESSON PLAN

Session No.	Planned Date	Content	Actual Date's	Mode of Delivery (Classroom/ Online)	Link for online content	Link for Course Content	Comments, if any by CI/HoD
1.		Module 1: The DevOps Culture and Infrastructure- What DevOps is · Dev vs Ops gap · DevOps culture and mindset · Benefits to an organization · DevOps team model		Classroom	https://youtu.be/2N-59wUIPVI?si=_9vI9CnhhF3CMwdj		
2.		Getting started with DevOps		Classroom	https://youtu.be/2N-59wUIPVI?si=_9vI9CnhhF3CMwdj https://youtu.be/2N-59wUIPVI?si=_9vI9CnhhF3CMwdj		

3.		Implementing CI/CD and continuous deployment		Classroom	https://www.youtube.com/watch?v=scEDHsr3APg https://youtu.be/72YTWEdKwyo?si=wQ0ztyOpN70ScxX8		
4.		Continuation of Implementing CI/CD and continuous deployment		Classroom	https://youtu.be/72YTWEdKwyo?si=wQ0ztyOpN70ScxX8 https://www.youtube.com/watch?v=HjXTSbXG1k8		
5.		Understanding IaC practices, The benefits of IaC		Classroom	https://www.youtube.com/watch?v=efRu7odnEQo https://youtu.be/mbcQai9iE48?si=Lye937uEcYW7yTVh		
6.		The evolution of the DevOps culture 2		Classroom	https://www.youtube.com/watch?v=LZj9P7Uvqd8		
7.		IaC languages and tools		Classroom	https://www.youtube.com/watch?v=IvY_o23ylzs		
8.		The IaC Topology		Classroom	https://www.youtube.com/watch?v=IvY_o23ylzs		

					tch?v=I9lKewvncX0 https://www.youtube.com/watch?v=K-4BF1_o5oo		
9.		Continuous Integration and Continuous Delivery: CI/CD principles		Classroom	https://www.youtube.com/watch?v=HjXTSbXG1k8		
10.		Module 2: Managing Your Source Code with Git-Getting started with Git		Classroom	https://www.youtube.com/watch?v=e9lnsKotSQ https://www.youtube.com/watch?v=a9u2yZvsqHA		
11.		Technical requirements		Classroom	https://www.youtube.com/watch?v=2ReR1YJrNOM		
12.		Overviewing Git and its principal command lines		Classroom	https://youtu.be/87ZqwoFeO88?si=E3stpKrYQz5u1k4A		
13.		Git configuration		Classroom	https://youtu.be/87ZqwoFeO88?si=E3stpKrYQz5u1k4A		
14.		Useful Git vocabulary		Classroom	https://www.youtube.com/watch?v=Uz_mTOQL9Tw		

15.		Git command lines		Classroom	https://youtu.be/SHdme1YdrAo?si=WqL2akq9gnBBUqhQ https://www.youtube.com/watch?v=HVsySz-h9r4 https://www.youtube.com/watch?v=PSJ63LULKHA		
16.		Continuation of Git command lines		Classroom	https://www.youtube.com/watch?v=HVsySz-h9r4 https://www.youtube.com/watch?v=rE2zRhZdjFU		
17.		Understanding the Git process and Gitflow pattern		Classroom	https://youtu.be/ilkhMPAiwl4?si=X413e4fqVR47QB-r		
18.		Branching strategy with Gitflow, Gitflow tools		Classroom	https://youtu.be/ilkhMPAiwl4?si=X413e4fqVR47QB-r		
19.		Module 3: Containerization and Development Automation-Containerizing Your Application with Docker- Technical requirements, Installing Docker, Registering on Docker Hub, Docker installation		Classroom	https://www.youtube.com/watch?v=C-bX86AgyiA		

20.		An overview of Docker's elements		Classroom	https://www.youtube.com/watch?v=90Y6YhmwFVs		
21.		Building and running a container on a local machine		Classroom	https://www.youtube.com/watch?v=eyNBflsqdBQ https://www.youtube.com/watch?v=0qotVMX-J5s		
22.		Continuation of Building and running a container on a local machine		Classroom	https://www.youtube.com/watch?v=Qv757wM_NQ0		
23.		Testing a container locally		Classroom	https://www.youtube.com/watch?v=sNg0bnMF_qY&t=209s		
24.		Continuation of Testing a container locally		Classroom	https://www.youtube.com/watch?v=sNg0bnMF_qY&t=209s https://www.youtube.com/watch?v=G2O5GEdZdPU		
25.		Pushing a Docker image to a private registry (ACR)		Classroom	https://www.youtube.com/watch?v=e3-rwiyz_wY&t=265s https://www.youtube.com/watch?v=sDxzgWIylZY		

26.		Deploying a container to ACI with a CI/CD pipeline		Classroom	https://www.youtube.com/watch?v=yjbzWq7bvNc		
27.		Module 4: A Brief Introduction to User Experience (UX) Design- A Brief Introduction to User Experience (UX) Design		Classroom	https://youtu.be/jLQauZP2e_4?s_i=Zcm4CAUrC6wJNTP- https://www.youtube.com/watch?v=ODpB9-MCa5s		
28.		What is User Experience (UX), UX Designer,		Classroom	https://youtu.be/jLQauZP2e_4?s_i=Zcm4CAUrC6wJNTP- https://youtu.be/80LM3-vwsDw?si=OMm8BQ_CNDoeWZS1		
29.		Methodology for UX		Classroom	https://youtu.be/jLQauZP2e_4?s_i=Zcm4CAUrC6wJNTP- https://youtu.be/80LM3-vwsDw?si=OMm8BQ_CNDoeWZS1		
30.		What is Design Thinking?,		Classroom	https://www.youtube.com/watch?v=HInixSID4gg		
31.		Design Thinking's Phases		Classroom	https://www.youtube.com/watch?v=HInixSI		

					D4gg https://www.youtube.com/watch?v=qyoZTUGzdGY		
32.		Design Thinking or 'Outside the Box' Thinking		Classroom	https://www.youtube.com/watch?v=r0VX-aU_T8 https://www.youtube.com/watch?v=SJXleXSglFQ		
33.		Design Thinking is an Essential Tool—and A Third Way		Classroom	https://www.youtube.com/watch?v=IDlhUpfnUtQ&list=PLuvAiVQM3SboAfV72wm9Vu8_bo5taOWuO		
34.		Design Thinking is an Iterative and Non-linear Process		Classroom	https://www.youtube.com/watch?v=IDlhUpfnUtQ&list=PLuvAiVQM3SboAfV72wm9Vu8_bo5taOWuO		
35		Module 5: The Factors that Influence User Experience- The 7 Factors that Influence User Experience		Classroom	https://www.youtube.com/watch?v=5aIQGQVoIf0 https://www.youtube.com/watch?v=IPdsFaM7HCs		

36.		Continuation of The 7 Factors that Influence User Experience		Classroom	https://www.youtube.com/watch?v=XADIFZbm3g4&t=409s		
37.		Why Does Usability Matter?		Classroom	https://www.youtube.com/watch?v=A8dlGqsAGXA		
38.		The 5 Characteristics of Usable Products		Classroom	https://www.youtube.com/watch?v=DYqRIqx-Elw		
39.		What is Interaction Design.		Classroom	https://youtu.be/X4ICTtzWlMk?si=g0j_Z2kMgFTjYaJO https://www.youtube.com/watch?v=j2c23lYUQb0		
40.		The 5 Dimensions of Interaction Design		Classroom	https://youtu.be/X4ICTtzWlMk?si=g0j_Z2kMgFTjYaJO https://www.youtube.com/watch?v=j2c23lYUQb0		
41		Introduction to Figma: Frames, Components and Styles		Classroom	https://www.youtube.com/watch?v=uQsyobT2Rv8		
42		Interactive Prototyping and User Flows		Classroom	https://www.youtube.com/watch?v=vkJGd33uOeo		

Textbook: (Maximum of 2 books) *In standard citation format*

1. Learning DevOps, 2nd Edition by Mikael Krief, published by Packt Publishing
2. Don't Make me think 2nd edition Steve Krug

Reference Books:

1. Scott Chacon and Ben Straub, "Pro Git", Apress, 2nd Edition, 2014.
2. Nigel Poulton, "Docker Deep Dive", Independently Published.
3. Jenifer Tidwell, "Designing Interfaces", O'Reilly Publications.
4. Figma Official Documentation.
5. GitHub Official Documentation

Online Resources:

1. Coursera - **Introduction to DevOps by IBM**
2. **Coursera**- UI / UX Design Specialization



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Evaluation pattern

BT Level	Test-1 (25 Marks)	Test-2 (25 Marks)	Project Learning (50 Marks)	SEE (100 Marks)
L1	-	-	-	-
L2	05	05	10	-
L3	15	10	10	25
L4	05	10	15	25
L5	-		15	-

LAB Evaluation

Component	Sub-Component	Evaluation Criteria	Marks
Lab Continuous Internal Evaluation (Lab CIE): 30 marks	Experiment Write up and + Execution	Correct implementation and completion of experiments	10
	Lab Record (Git Repository)	Timely updating and documentation of their Git Repo	10
	Viva (Lab Quiz)	Quiz	10

Assessment Rubric for Project learning (50 Marks)

Sl.No	Component	Marks
1	Problem Definition	10
2	Literature Survey and Review	10
3	Model Design & Implementation	20
4	Presentation & Technical Report	10
	Total	50



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Course Instructor:

Course Title: Collaborative Development and DevOps

LTP: 3:0:2

SEC

Course Code: 1BCS303

Semester: VI

LAB LESSON PLAN

Session No.	Planned Date	Content	Actual Date's	Mode of Delivery (Classroom/ Online)	Link for online content	Link for Course Content	Comments, if any by CI/HoD
1.		Installation and configuration of Git. Create a local repository and perform basic Git operations.		LAB	https://www.youtube.com/watch?v=9BbS46HwjBM		
2.		Perform Git operations: clone, push, pull, fetch and repository synchronization.		LAB	https://www.youtube.com/watch?v=Rxn0tBNyQBw		
3.		Implement branching, merging, rebasing and resolve merge conflicts.		LAB	https://www.youtube.com/watch?v=nMxefyNCNX0		
4.		Create GitHub repositories, manage issues, forks, pull requests and code reviews.		LAB	https://www.youtube.com/watch?v=oxgk6J5yL-Y		
5.		Collaborate on a team project using GitHub repositories and branching strategies.		LAB	https://www.youtube.com/watch?v=wDRodui98&vl=en		

6.		Create Continuous Integration workflows using GitHub Actions		LAB	https://www.youtube.com/watch?v=0Pbxplao_EU		
7.		Automate build and testing processes using GitHub Actions workflows.		LAB	https://www.youtube.com/watch?v=R8_veQiYBjI		
8.		Create Docker images and containers for simple applications.		LAB	https://www.youtube.com/watch?v=fqMOX6JJhGo		
9.		Write Dockerfiles and build custom containerized applications		LAB	https://www.youtube.com/watch?v=3c-iBn73dDE		
10.		Develop multi-container applications using Docker Compose.		LAB	https://www.youtube.com/watch?v=HG6yIjZapSA		
11		Containerize and execute a Java or Python application using Docker.		LAB	https://www.youtube.com/watch?v=3c-iBn73dDE		
12		Design user personas, user journeys, and wireframes for an application		LAB	http://youtube.com/watch?v=jwCmIBJ8Jtc		
13		Design high-fidelity UI prototypes and reusable components using Figma.		LAB	https://www.youtube.com/watch?v=FTFaQWZBqQ8		
14		Mini Project: Develop a team-based application incorporating GitHub collaboration, CI workflow, Docker containerization, UX research, and UI prototyping.		LAB	https://www.youtube.com/watch?v=9pZ2xmsSDdo		

